

Mehmet Baha Dursun

Robotics engineer building systems across autonomous navigation, surgical robotics, and vision-language-action learning. Experienced in deploying ROS/ROS 2 systems on commercial robots and translating learning-based control from research code to physical hardware.

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EXPERIENCE

Robotics Engineer

Nov 2023 — Present

Saha Robotik · Istanbul / Denmark

- Lead localization and mapping work for 200+ autonomous service robots operating in commercial environments.
- Managed the navigation stack for a 24/7 airport guide robot, covering a 187,000 m² terminal and at least 25 km per day.
- Developing the SMR10 + bimanual OpenArm robot-learning platform: ROS 2 control and safety, LeRobot v3 data pipelines, MuJoCo simulation, and OpenPI/π0.5 serving on Jetson Orin.

Student Research Assistant

Oct 2025 — Present

University of Southern Denmark · VP AURI Lab

- Develop and validate vision-guided control for the da Vinci Research Kit, spanning synchronized demonstrations, learning, diagnostics, and ROS 1 robot evaluation.
- Built a bimanual Diffusion Policy pipeline with DINOv2 perception, 20D anchored-relative actions, LeRobot v3 data, and live/offline policy-grounding diagnostics.

Team Lead → Mentor

2021 — Present

KOU-Mekatronom Autonomous Vehicle Team

Led 20 students across Robotaxi and Bosch Future Mobility Challenge; global top-24 BFMC finalist and two sixth-place Robotaxi finishes.

SELECTED RESEARCH & ENGINEERING

Vision Encoders for a Fixed dVRK Diffusion Policy

May 2026

Compared DINOv2 ViT-S/14 and ResNet34 on 84 real suturing demonstrations (43,890 frames). DINOv2 reached lower comparable validation loss (0.00268 vs. 0.00436), localized the needle more selectively, and reached 0.995 measured vision share with state dropout. Individual SDU course research report.

KOU-Mekatronom Autonomous Vehicle Stack

2022 — 2025

Integrated NMPC with CasADi, visual-inertial localization, sensor fusion, obstacle detection, planning, and competition deployment in ROS, C++, Python, MATLAB, OpenCV, and YOLO.

EDUCATION

University of Southern Denmark

MSc studies in Robotics & Autonomous Systems

2025 — Present · Full scholarship

Kocaeli University

BSc, Mechatronics Engineering

2019 — 2024 · GPA 3.2 / 4.0

TECHNICAL SKILLS

Programming

Python, C/C++, MATLAB

Robot learning

PyTorch, Diffusion Policy, DINOv2, VLA, LeRobot, OpenPI

Robotics

ROS / ROS 2, control, SLAM, localization, sensor fusion, planning

Simulation & CV

MuJoCo, OpenCV, YOLO, calibration

Systems

Linux, Docker, Git, Jetson Orin

RESEARCH INTERESTS

Robot learning, embodied AI, surgical robotics, visual representations, sim-to-real transfer, and reliable autonomy.

LANGUAGES

Turkish · Native

English · Professional working proficiency (IELTS 6.5)